Inequality

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Narrative arc

Concepts

↓

Individuals

↓

Populations

↓

Societies

In a model world where personal wealth follows random multiplicative growth.

Model world

Geometric Brownian Motion (GBM) for wealth $x(t)$,

$$dx = x\left(\mu dt + \sigma dW\right),$$

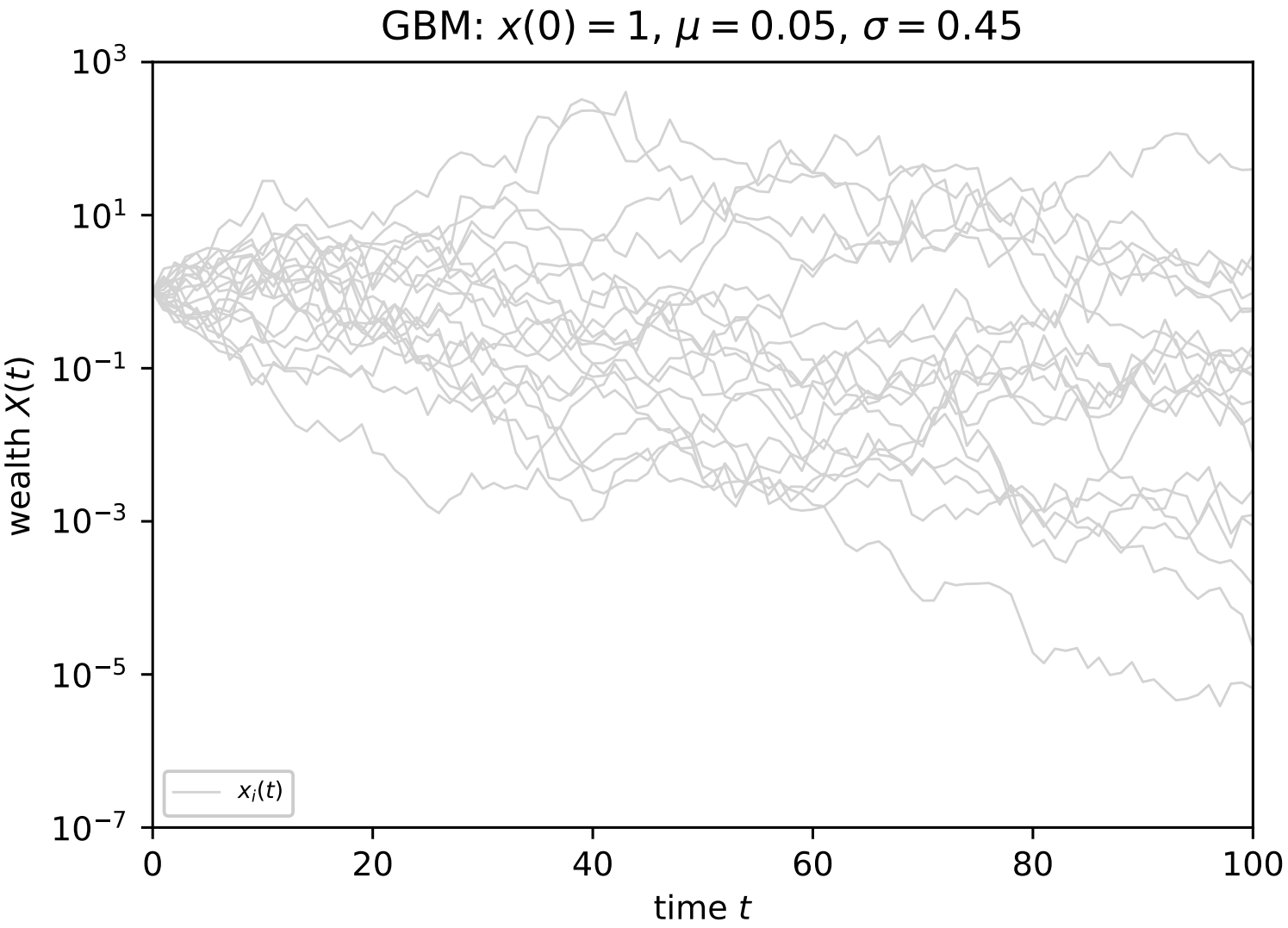
where:

- drift μ – deterministic growth;
- volatility σ – fluctuation amplitude;
- Wiener increment $dW \sim \mathcal{N}(0, dt)$ – randomness.

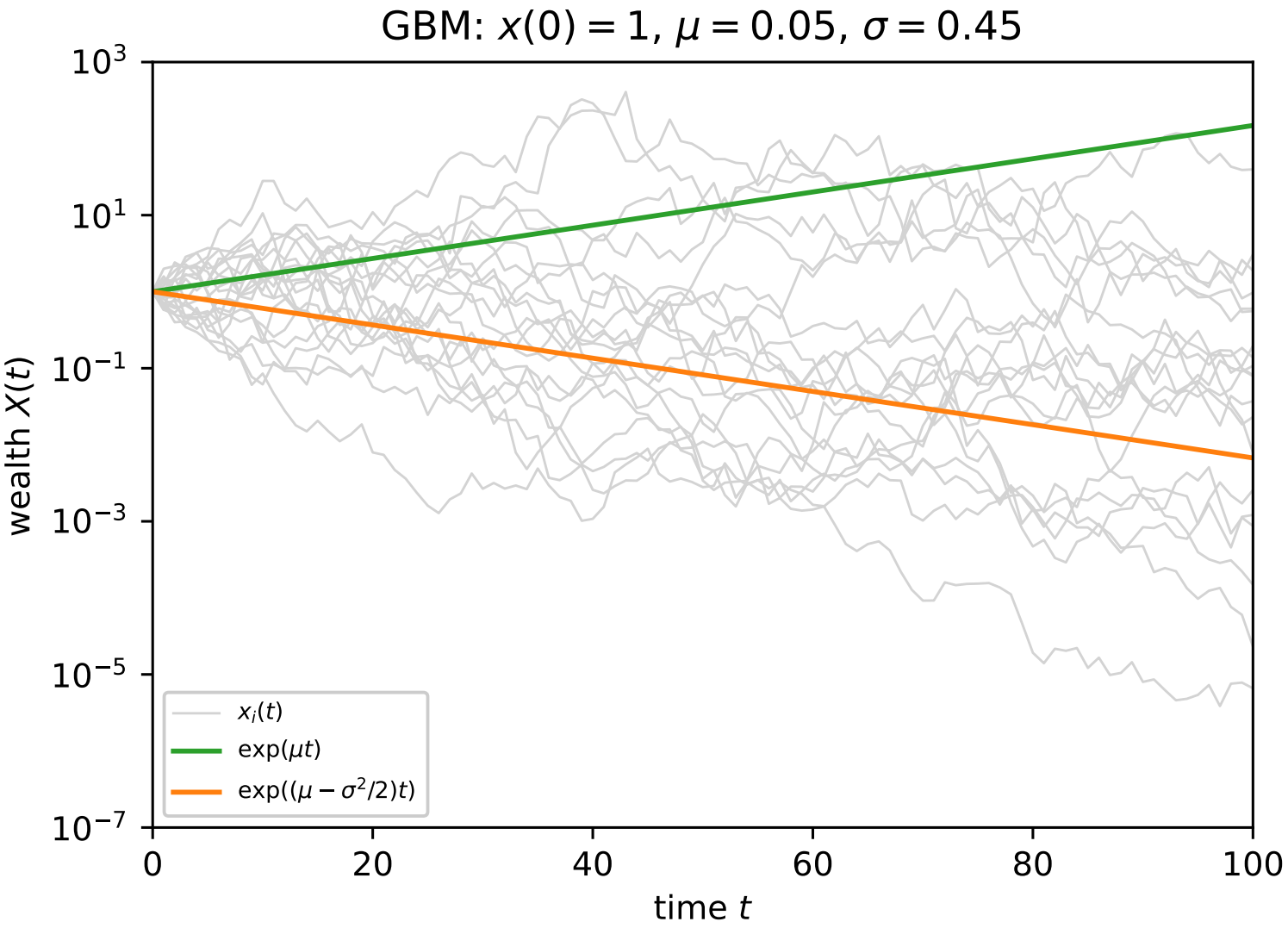
Generalisation of repeated coin toss.

Our model world contains N independent trajectories, $x_i(t)$.

Model world


GBM: $x(0) = 1$, $\mu = 0.05$, $\sigma = 0.45$
10^3
10^1
10^{-1}
10^{-3}
10^{-5}
10^{-7}
wealth $X(t)$
0
20
40
60
80
100
time t
$x_i(t)$

Model world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10^3
10^1
10^{-1}
10^{-3}
10^{-5}
10^{-7}
wealth $X(t)$
0
20
40
60
80
100
time t
$x_i(t)$
$\exp(\mu t)$
$\exp((\mu - \sigma^2/2)t)$

Lognormal distribution

8.1.1 Lognormal distribution

Recall from Sec. 4.4 that the SDE for GBM,

$$dx = x(\mu dt + \sigma dW), \tag{8.1}$$

has solution

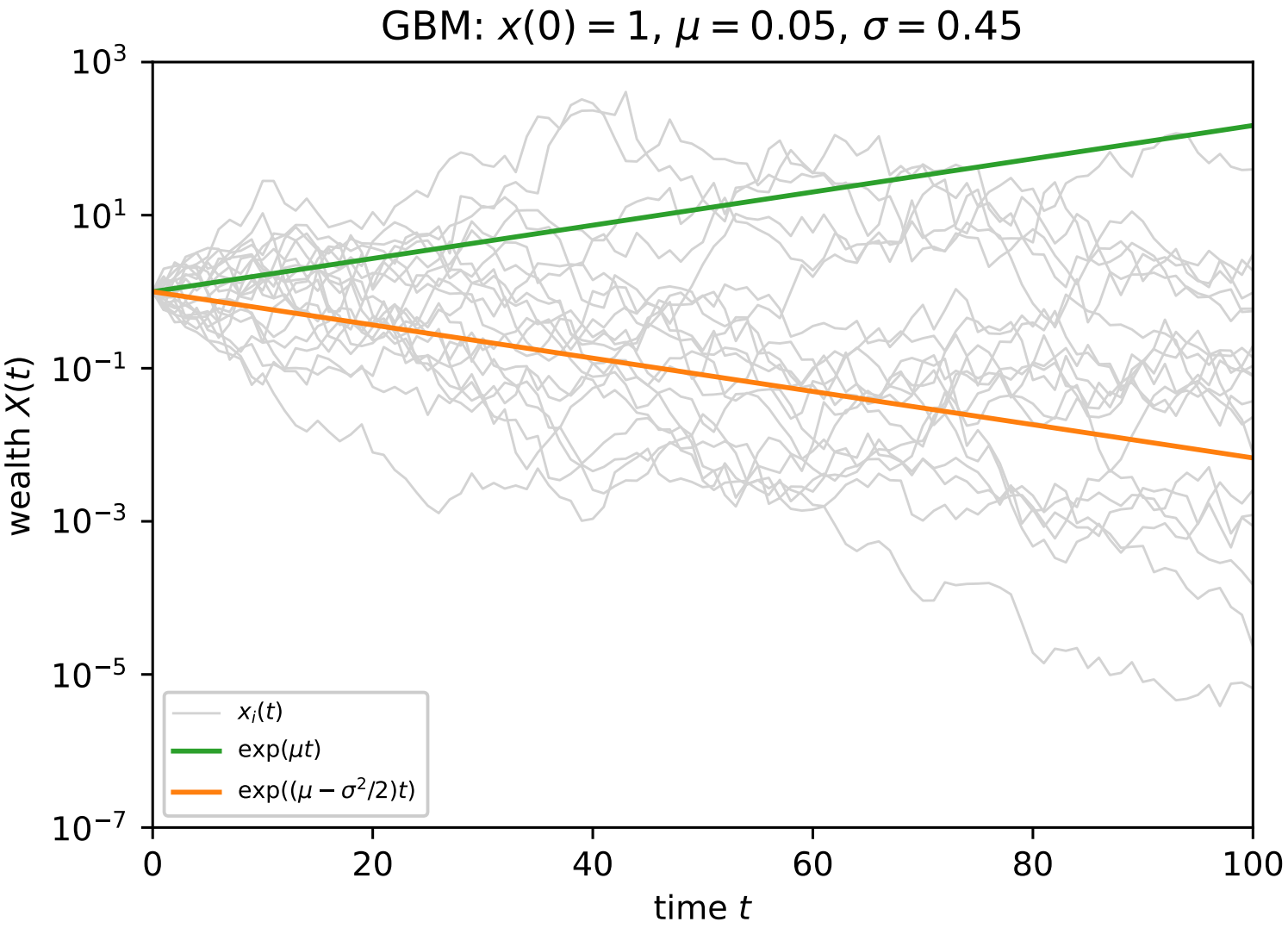
$$x(t) = x(0) \exp\left[\left(\mu - \frac{\sigma^2}{2}\right) t + \sigma W(t)\right], \tag{8.2}$$

which, at a given time t, can be viewed as a lognormally-distributed random variable with time-dependent parameters,

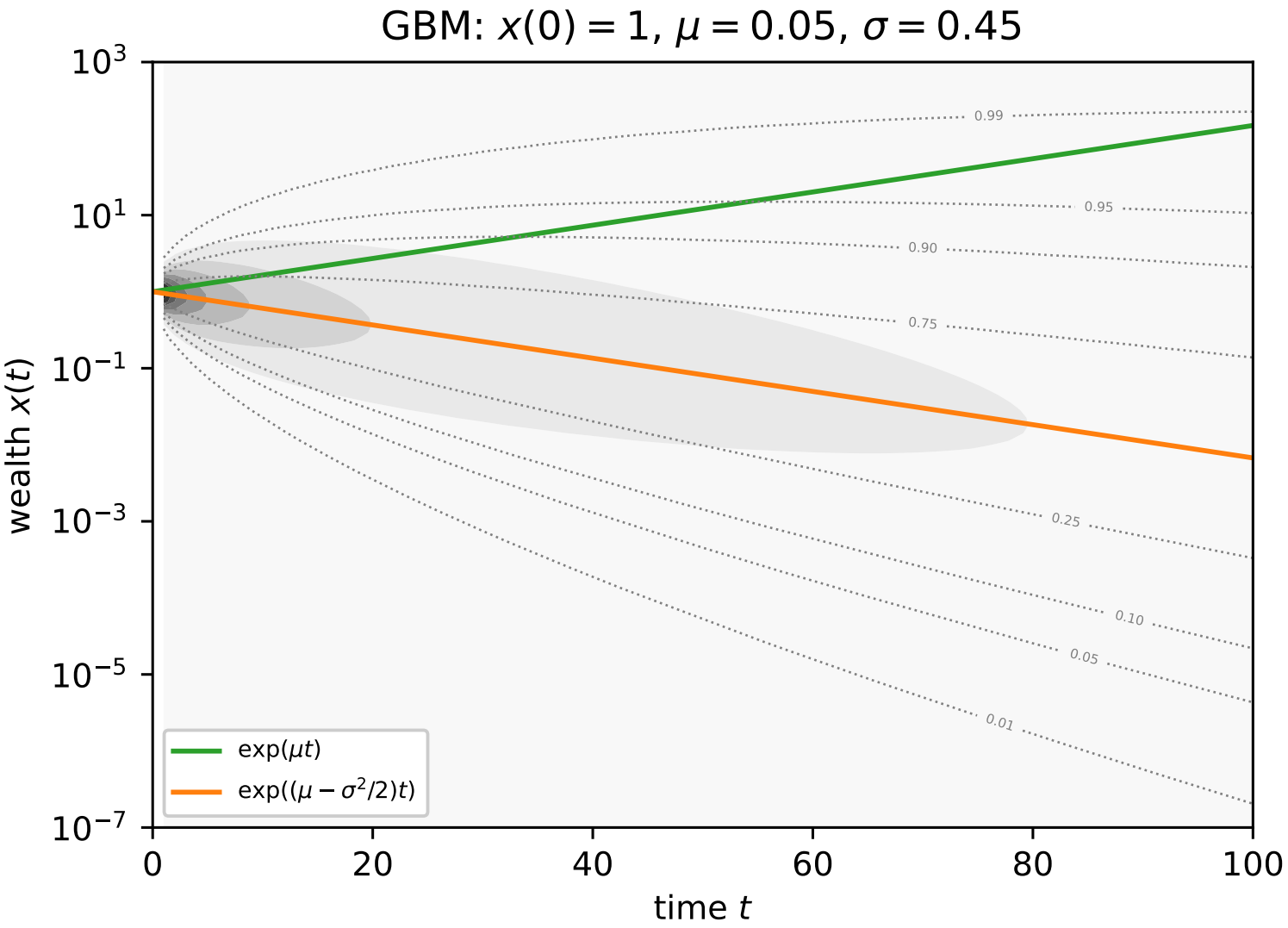
$$\ln x(t) \sim \mathcal{N}\left(\ln x(0) + \left(\mu - \frac{\sigma^2}{2}\right) t, \sigma^2 t\right). \tag{8.3}$$

In our model population, we will assume that all individuals have the same starting wealth, $x(0)$.

Lognormal distribution



Lognormal distribution



Two growth rates

Two ways to quantify the "average" wealth growth rate in our model population:

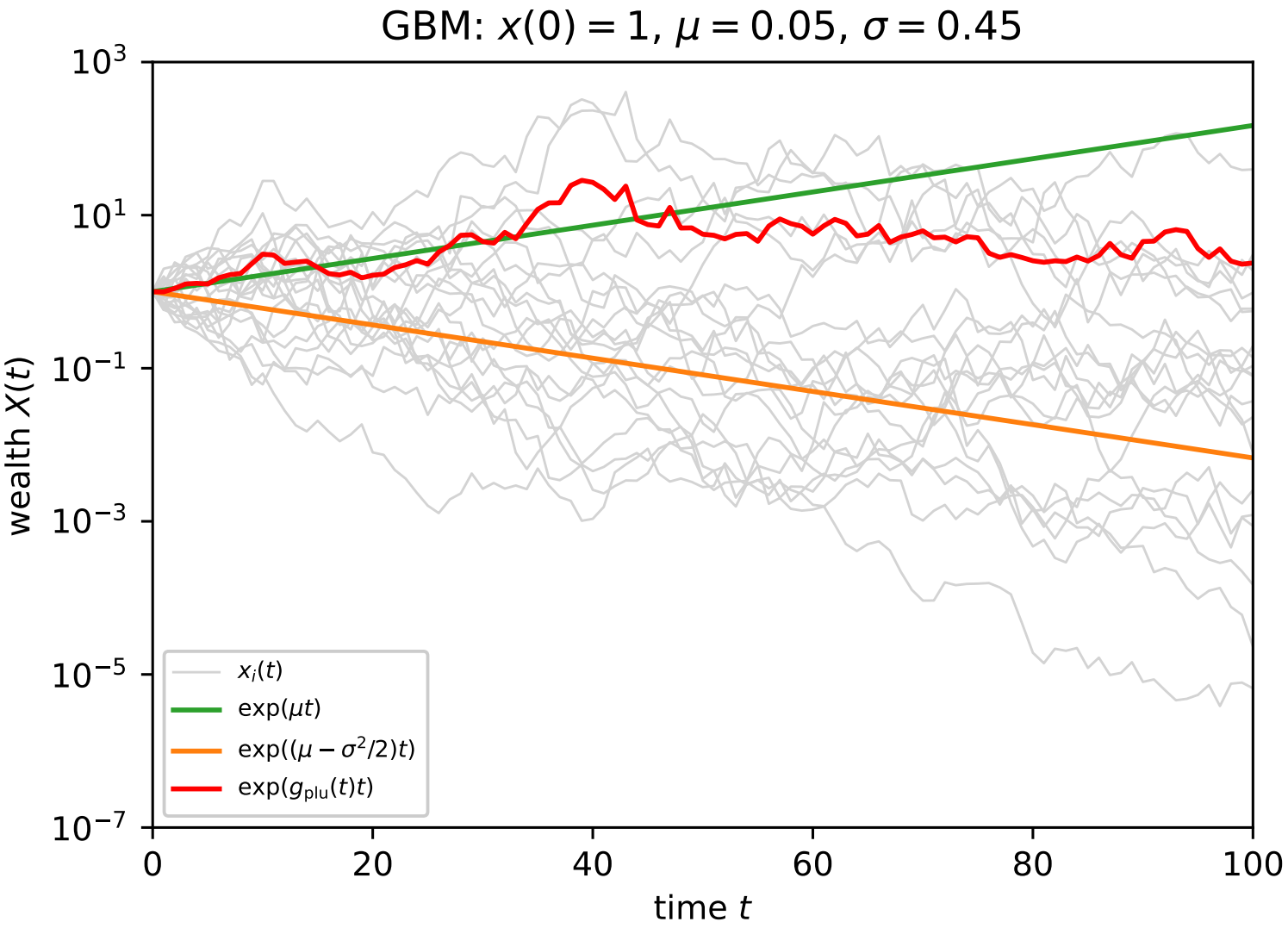
1. Plutocratic growth rate

Find the population average wealth, $\langle x \rangle_N$, and compute its ergodic growth rate:

$$g_{\text{plu}} \equiv \frac{\Delta \ln \langle x \rangle_N}{\Delta t}.$$

"Plutocratic" because each individual makes a wealth-weighted contribution to the average.

Two growth rates



Two growth rates

Two ways to quantify "average" wealth growth rate in our model population:

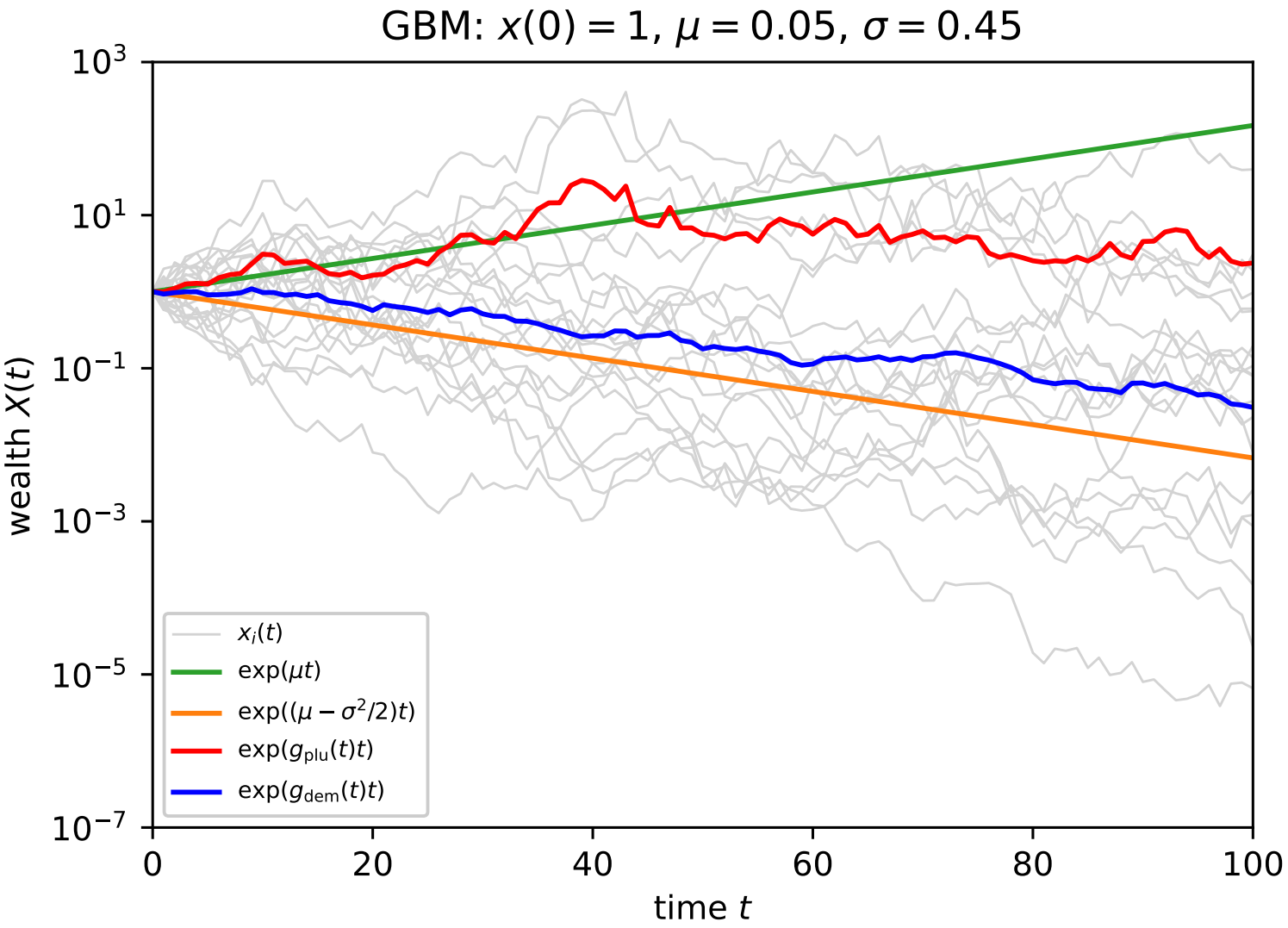
2. Democratic growth rate

Find the ergodic growth rate of each wealth, $\frac{\Delta \ln x_i}{\Delta t}$, and compute the population average:

$$g_{\text{dem}} \equiv \left\langle \frac{\Delta \ln x}{\Delta t} \right\rangle_N .$$

"Democratic" because each individual makes an equal contribution to the average.

Two growth rates



Ergodicity

For finite N, both g_{plu} and g_{dem} are random variables.

We can replace them with model properties as $N \to \infty$:

$$g_{\text{plu}} = \frac{\Delta \ln \langle x \rangle_N}{\Delta t} \quad \to g_{\langle\,\rangle} = \mu;$$

$$g_{\text{dem}} = \left\langle \frac{\Delta \ln x}{\Delta t} \right\rangle_N \quad \to \bar{g} = \mu - \frac{\sigma^2}{2} \text{ (ergodicity).}$$

Good approximations for "large" model populations.

Discussion

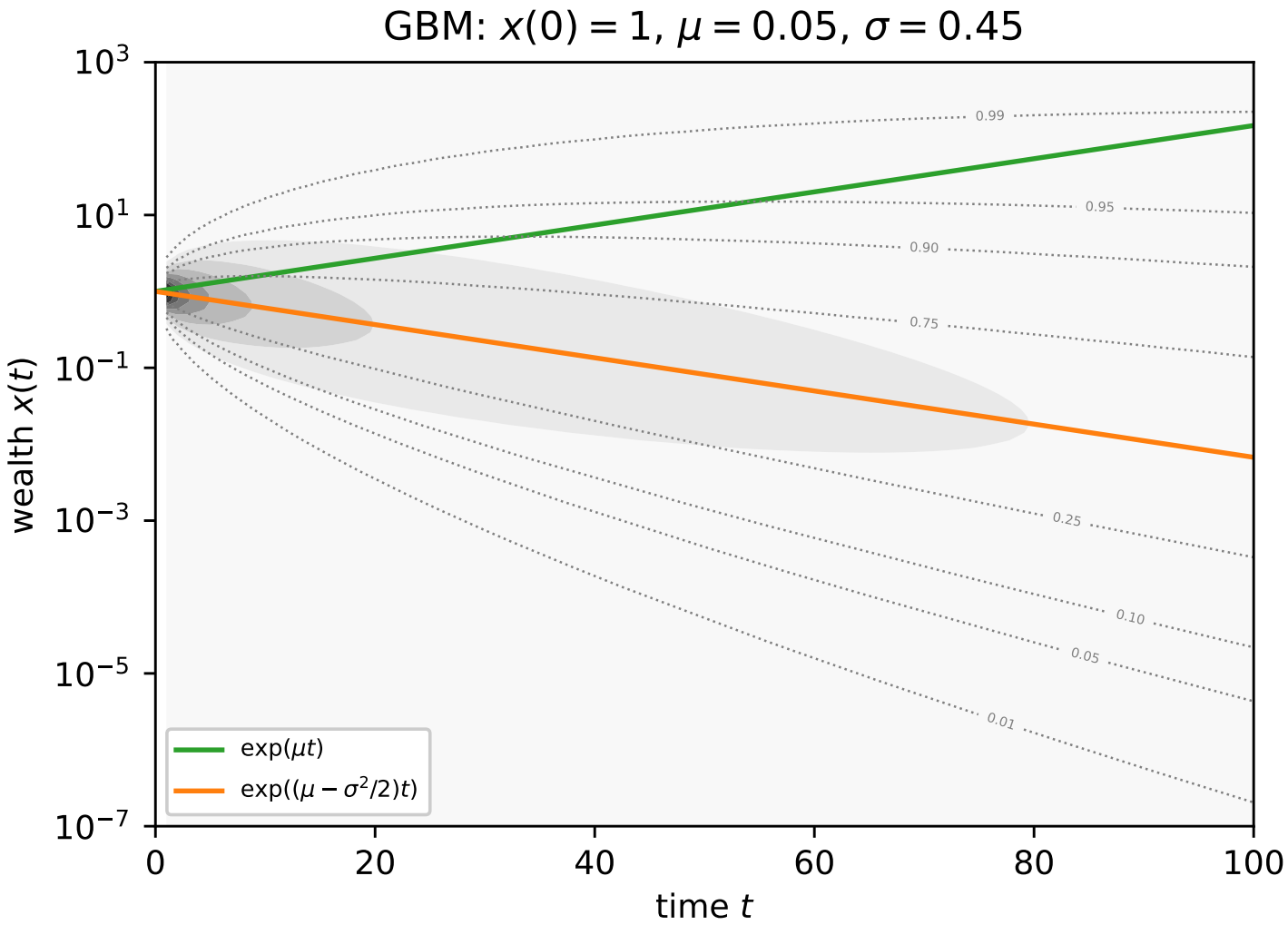
Jeff Bezos in the Library (with the Extreme Wealth).

Gross Domestic Product – government vs. the people?

"The two growth rates of the economy"
Yonatan Berman, OP, AA (2020)
https://arxiv.org/abs/2009.10451.

Inequality

"A whole menagerie of inequality measures in economics."



Recipe for inequality metric à la EE

1. Note that: if $g_{\text{plu}} > g_{\text{dem}}$, then inequality is increasing; and if $g_{\text{plu}} < g_{\text{dem}}$, then inequality is decreasing.

2. Therefore, define inequality, $J(t)$, as the thing that grows at the **difference** between the two rates:

$$\frac{dJ}{dt} = g_{\text{plu}} - g_{\text{dem}}.$$

3. Integrate from time 0 (equality) to t to get:

$$J(t) = \ln \langle x(t) \rangle_N - \langle \ln x(t) \rangle_N .$$

Inequality

J is the **mean logarithmic deviation** or **Theil's L index**.

For a large population, $N \to \infty$, following GBM,

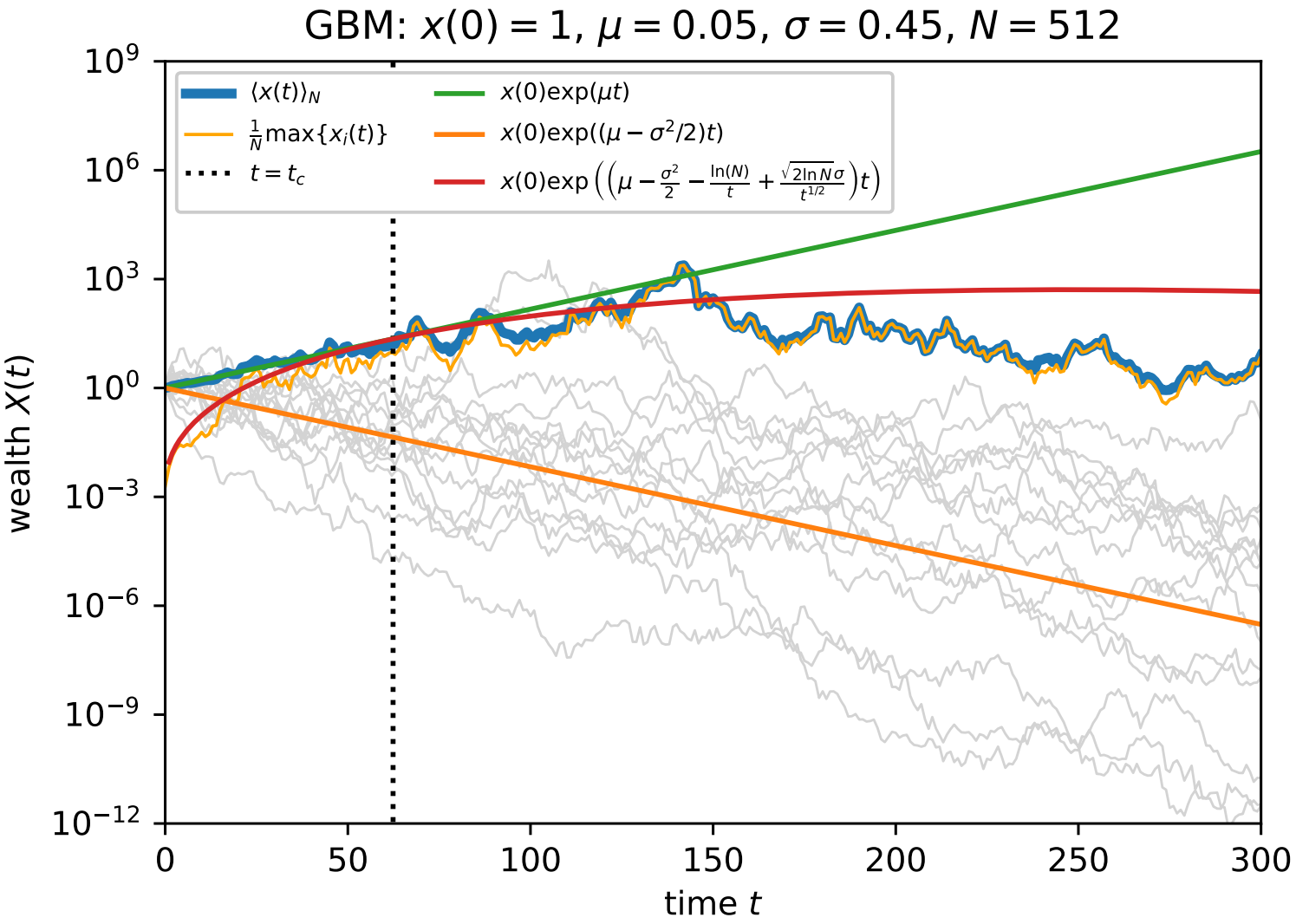
$$\frac{dJ}{dt} = g_{\langle\,\rangle} - \bar{g} = \frac{\sigma^2}{2} \quad \Rightarrow \quad J(t) = \frac{\sigma^2 t}{2},$$

so inequality grows without bound.

For general wealth dynamics with ergodicity transformation, $v(x)$, wealth is v-normally distributed and

$$J_v(t) = v\left(\langle x(t)\rangle_N\right) - \langle v\left(x(t)\right)\rangle_N .$$

Finite populations



Thank you.